

Assignment No. 1

Asad Ullah

Electrical Ai

Semester 2nd

Section B

Registration BF25NWELE0686

```
#include <iostream>                                // Library for input/output
using namespace std;                               // Allows using cout, cin without std::

// Built in Function to convert grade (A, B, etc.) into grade points
double getGradePoint(char grade) { // Function starts
    if (grade == 'A') return 4.0; // If grade is A, return 4.0
    else if (grade == 'B') return 3.0; // If grade is B, return 3.0
    else if (grade == 'C') return 2.0; // If grade is C, return 2.0
    else if (grade == 'D') return 1.0; // If grade is D, return 1.0
    else return 0.0; // For F or invalid grade, return 0
}

int main() {                                       // Main function starts

    // i. Declare variables
    double voltage, current, resistance;          // Declaring three double variables

    // ii. Display message
    cout << "Welcome to OOPs in C++!" << endl;  // Printing welcome message
```

// iii. Read age

```
int age; // Declare integer variable for age
cout << "Enter your age: "; // Ask user to enter age
cin >> age; // Take input from user
```

// iv. Temperature check

```
double temperature; // Declare temperature variable
cout << "Enter temperature: "; // Ask for temperature
cin >> temperature; // Input temperature
if (temperature > 100) // Check if temperature is greater than 100
    cout << "Overheating detected!\n"; // Print warning
```

// v. Area of circle

```
double r, area; // Declare radius and area
cout << "Enter radius: "; // Ask for radius
cin >> r; // Input radius
area = 3.14159 * r * r; // Calculate area using formula  $\pi r^2$ 
cout << "Area of circle: " << area; // Display area
```

// vi. Print x, y, z

```
int x, y, z; // Declare three integers
cout << "Enter x, y, z: "; // Ask user
cin >> x >> y >> z; // Input values
cout << x << ", " << y << ", " << z << endl; // Print values separated by comma
```

// vii. Sum of voltage and current

```
cout << "Enter voltage and current: "; // Ask user
cin >> voltage >> current; // Input voltage and current
double total = voltage + current; // Calculate sum
cout << "Total: " << total << endl; // Display result
```

// viii. Division check

```
double numerator, denominator;  
cout << "Enter numerator and denominator: ";  
cin >> numerator >> denominator;  
if (denominator == 0)  
    cout << "Error: Division by zero" << endl;
```

```
// Declare variables  
// Ask user  
// Input values  
// Check if denominator is zero  
// Display error
```

// ix. Increment count

```
int count;  
cout << "Enter count: ";  
cin >> count; // Input value  
cout << "After increment: " << ++count << endl;
```

```
// Declare count  
// Ask user  
  
// Increment and print
```

// x. Read three floats

```
float a, b, c;  
cout << "Enter three numbers: ";  
cin >> a >> b >> c;
```

```
// Declare three float variables  
// Ask user  
// Input values
```

// xi. Series resistance

```
double r1, r2, r3;  
cout << "Enter three resistances (series): ";  
cin >> r1 >> r2 >> r3; // Input values  
cout << "Total Series Resistance: " << (r1 + r2 + r3);
```

```
// Declare resistances  
// Ask user  
  
// Formula  $R = r1 + r2 + r3$ 
```

// xii. Parallel resistance

```
cout << "Enter three resistances (parallel): ";  
cin >> r1 >> r2 >> r3; // Input values  
cout << "Total Parallel Resistance: " << 1 / (1/r1 + 1/r2 + 1/r3) << endl;
```

```
// Ask user  
  
  
// Formula for parallel
```

// xiii. Voltage division

```
double V, R1, R2;  
cout << "Enter V, R1, R2: ";  
cin >> V >> R1 >> R2;  
cout << "Voltage across R2: " << V * (R2 / (R1 + R2)) << endl;
```

// Declare variables

// Ask user

// Input values

// Voltage division formula

// xiv. Current division

```
double I;  
cout << "Enter total current I, R1, R2: ";  
cin >> I >> R1 >> R2; // Input values  
cout << "Current through R2: " << I * (R1 / (R1 + R2)) << endl;
```

// Declare current

// Ask user

// Current division formula



// xv. GPA calculation using alphabet grades

```
char g1, g2, g3;                                     // Declare grade variables (A, B,
                                                    etc.)
double ch1, ch2, ch3;                                // Declare credit hours

cout << "Enter grades (A/B/C/D/F) and credit hours:\n"; // Ask user

cout << "Subject 1: ";                               // Prompt for subject 1
cin >> g1 >> ch1;                                     // Input grade and credit hours

cout << "Subject 2: ";                               // Prompt for subject 2
cin >> g2 >> ch2;                                     // Input grade and credit hours

cout << "Subject 3: ";                               // Prompt for subject 3
cin >> g3 >> ch3;                                     // Input grade and credit hours

double gp1 = getGradePoint(g1);                     // Convert grade to points
double gp2 = getGradePoint(g2);                     // Convert grade to points
double gp3 = getGradePoint(g3);                     // Convert grade to points

double gpa = (gp1*ch1 + gp2*ch2 + gp3*ch3) / (ch1 + ch2 + ch3);
                                                    // GPA formula

cout << "Semester GPA: " << gpa << endl;             // Display GPA

return 0;                                           // End of program
}
```